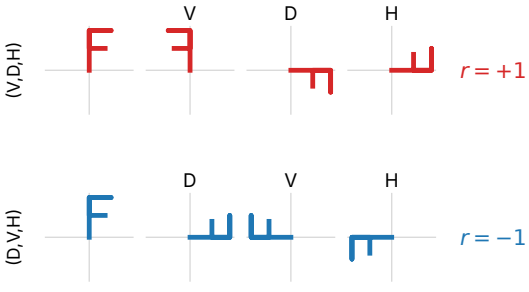


(b) scalar Q, order-blind

histories (V, D) and (D, V) both observed as $\sigma = \{V : 1, D : 1\}$, action H

one number $f(\sigma, H)$ for both; targets $+1$ and -1
$\Rightarrow$ error ≥ 1 (Prop. 8)

(c) operator Q, compositional backup

$Q(s, a) = \mathcal{K}_a^\dagger(R + \gamma \mathbb{E}V)$
unrolls to $A_{a_1} \cdots A_{a_T} W$ (Cor. 1)
one $N \times N$ operator per (s, a)

value $= \tau(X_h Q(H))$, $Q(H) = A_H W$:
$X_{VD} = A_V A_D \Rightarrow +1$, $X_{DV} = A_D A_V \Rightarrow -1$
exact for every history (Prop. 1)

(d) the backup runs backwards and composes

$Q_0(V) = \mathcal{K}_V^\dagger(Q_1)$
$= A_V A_D A_H W$

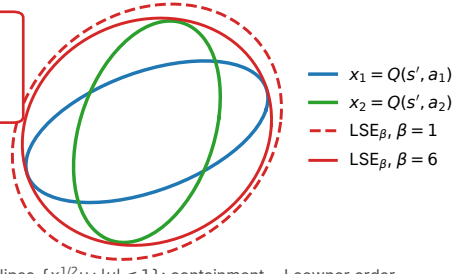
$Q_1(D) = \mathcal{K}_D^\dagger(Q_2)$
$= A_D A_H W$

$Q_2(H) = \mathcal{K}_H^\dagger(W)$
$= A_H W$

readout at $X_0 = I$: $\tau(Q_0(V)) = \frac{1}{N}\mathrm{Tr}(W A_V A_D A_H) = +1$; after history h: $\tau(X_h Q_{|h|})$

memory: $|\mathcal{S}|K$ operators (N^2 numbers each), for every horizon
vs. K^T entries for an exact history table (Prop. 9)

(e) control: no operator max; LSE is an upper bound



ellipse $\{x^{1/2}u : |u| \leq 1\}$; containment = Loewner order